

Microstimulation: Principles, Techniques, and Approaches to Somatosensory Neuroprosthesis

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ABSTRACT: The power of movement of electrically charged particles has been used to alleviate an array of illnesses and help control some human body parts. Microstimulation, the electrical current-driven excitation of neural elements, is now being aimed at brain-machine interfaces (BMIs), brain-controlled external devices that improve quality of life for people such as those who have lost the ability to use their limbs. This effort is motivated by behavioral experiments that indicate a direct link between microstimulation-induced sensory experience and behavior, pointing to the possibility of optimizing and controlling the outputs of BMIs. Several laboratories have focused on using electrical stimulation to return somatosensory feedback from prosthetic limbs directly to the user's central nervous system. However, the difficulty of the problem has led to limited success thus far, and there is a need for a better understanding of the basic principles of neural microstimulation. This article provides a review of the available literature and some recent work at Downstate Medical Center and Columbia University on microstimulation of the primate and rodent somatosensory (S1) cortex and the ventral posterolateral thalamus. It is aimed at contributing to the existing knowledge base to generate good behavioral responses and effective, BMI-appropriate somatosensory feedback. In general, the threshold for the particular brain tissue in response to current-amplitude has to be determined by rigorous experimentation. For consistently reproducible results, hardware and thresholds for microstimulation have to be specified. In addition, effects on motor functions, including unwanted side effects in response to the microstimulation of brain tissue, must be examined to take the field from bench to bedside.

KEY WORDS: microstimulation, somatosensory neuroprosthesis, BMI, sensory feedback

ABBREVIATIONS: BMI, brain-machine interface; C-C, cortico-cortical; CNS, central nervous system; cVPL, caudal ventral posterolateral thalamus; DCN, dorsal column nuclei; LFP, local field potential; LGC, linear Granger causality; M1, primary motor cortex; MRI, magnetic resonance imaging; mVPL, middle ventral posterolateral thalamus; PSTH, peristimulus time histogram; QA, quickly adapting; rVPL, rostral ventral posterolateral thalamus; S1, primary somatosensory cortex; SA, slowly adapting; SSNP, somatosensory neuroprosthesis; TDT, Tucker-Davis Technologies, Inc.; Vc, ventral caudal; VL, ventrolateral; VP, ventroposterior nucleus; VPL, ventral posterolateral thalamus.

I. INTRODUCTION

Tissue stimulation through electrical means has been studied for more than 200 years,¹ and serious consideration has been given to its use as a permanent substitute for impaired body function. For example, one of the ancient suggested remedies for patients suffering from ailments such as gout or headache was to touch electric fish in the hope that its powerful jolt might cure them.²

Although the use of electrical stimulation of the auditory nerve to replace lost hearing^{1,2} is a singular major step forward, this approach has affected our

lives in many ways. It has helped in the treatment of deep brain structures to control various motor and emotional deficits,^{3–11} restore vision,^{12–18} relieve pain,^{19–24} control seizures,²⁵ coordinate the movement of paralyzed limbs,²⁶ heal bone,^{27–34} restore gait and posture,^{35–37} control bladder,^{38–41} and correct impotence.^{42,43} Devices that use the same approach, such as pacemakers⁴⁴ and implantable cardioverter defibrillators,^{45,46} have gained acceptance in various therapeutic protocols and rehabilitation.

These efforts were aided by pioneer researchers such as Dr. Penfield and his team, who provided the scientific community with functional maps of the

brain through stimulation of various areas of the human cortex. A somatotopic representation of body organs that, when stimulated, elicit unique behavioral responses is well documented. Such investigations have made it possible to carefully stimulate chosen brain parts to control some ailments or to pinpoint the locus of an illness,^{47–49} for instance, an epileptic focus.^{25,50–54}

These studies provided an impetus to conduct research through direct electrical stimulation of the brain to control external devices (brain–machine interfaces [BMIs]), which aims at replacing functions, for instance, those lost by amputated or paralyzed limbs. There are some encouraging results,^{55,56} but all-inclusive studies that compare thresholds for task performance, conscious detection, and observable firing-rate modulation are still lacking. Classification of highly reliable and concrete limits that can assist us in using the full potential of microstimulation for functional restoration is desired.

Several questions need to be answered to enrich our knowledge base. Since the first comprehensive report of brain microstimulation by Ranck,⁵⁷ few papers have provided guiding principles to perform microstimulation.^{58–61} These recommendations came from experiments that used high-frequency cortical microstimulation. However, parameters of the successful stimulation have not been clearly established. There is a need for discussion on avoiding long-term side effects, such as epileptic seizures,^{62–64} and on the ethical issues of microstimulation.⁶⁵ It is important to note that the aforementioned microstimulation advances, and future approaches, mostly rely on the subject locally controlling the electrical current (i.e., the intensity and frequency) through self-stimulation. However, self-stimulation has long been shown to be important in behavior modification studies and maintenance of operant responses.^{60,66–70} Hence, although it is not the focus of this paper, it is appropriate to remark that cautionary checks should be built in.

Low-frequency microstimulations with a few stimulus parameter combinations can create neural responses with a complex pattern, as the data presented here and elsewhere demonstrate. This also illustrates that simple measures such as excitation

levels and how they spread can reliably predict how unit and population responses are modulated. Based on the goal of any given experiment, these parameters can be tailored to create specific neural responses and avoid negative consequences.

Answering complex questions about sensation and perception is generally not the goal of these approaches. Those issues have to be addressed with careful projects in humans where subjective evaluations can be made. Moreover, multichannel and optimized microstimulation can be used for further refining the desired response. As a starting point, there is a need for continuing discussion of the central ideas, including thresholds for observable modulation and major tractable impediments.

To describe the current status of somatosensory neuroprosthesis (SSNP) strategies and best organize these thoughts, three general aims can be hypothesized and tested:

1. To compare and contrast somatosensory neural responses evoked by natural touch and those evoked by cortical and thalamic microstimulation for a neural prosthesis. A significant difference in neural responses between natural and microstimulation paradigms is expected.
2. To find which brain region—the ventral posterolateral thalamus (VPL) or primary somatosensory cortex (S1)—best demonstrates the possibility of using microstimulation to create cortical neural responses that mimic those observed during natural touch stimulation. A significant difference in neural responses between VPL and S1 microstimulation paradigms is postulated.
3. To investigate how an electrical stimulus affects the natural dynamics of different brain areas involved and to find differences in functional connectivity. Toward this aim, differences in functional connectivity are hypothesized, but not to the extent that microstimulation has a long-lasting and deleterious effect, rendering its use for both rehabilitation and BMI applications impractical.

II. TISSUE ELECTRICAL STIMULATION: SOMATOSENSATION AND NEUROPROSTHESIS

A few encouraging results have led to the belief that the sensory perception of electrical impulses in the somatosensory thalamus,^{37,71–74} as well as in the cortex,^{75–78} has the potential to produce patterns of cortical population activity that are similar to those observed during natural touch to the arm, and to act as a cue to guide rodents and primates in navigation and reaching movements^{77,79,80} and active sensing.^{75,81}

Some methods, including single-channel microstimulation, have been successfully used to guide behavioral task performance in rats and monkeys.^{75,76,82–89} Research indicates that artificial devices such as robotic arms can be sensitized to give the user vital information about the external environment.⁹⁰

Understandably, before trying to mimic natural inputs, one has to be aware of how they are computed and what their characteristic features are. Experimentally, the process of deciding on the intensity, frequency, and so on of natural touch inputs into peripheral organs and measuring neural responses is relatively straightforward. One can test mechanical devices and get feedback from human subjects and observe neural responses with little or no pain or long-lasting adverse effects. Of course, this is without getting into neurophilosophical questions of perception and sensation. By contrast, the issue of injecting current into brain tissue is a much more complicated and potentially harmful process that requires a careful approach.

Nevertheless, it is now agreed that both humans and animals can discriminate signals delivered to sensory areas of their brains, including via the use of electrical microstimulation. It is thus proposed that neuroprosthetic devices with an artificial sensory channel can be made, but no one has yet provided suitable microstimulation protocols to test such a hypothesis.⁷⁷

The current candidate for SSNP feedback is cortical microstimulation. Its safety considerations and success limits, however, are not completely understood. Three main concerns call for further investigations:

1. There are direct connections between the motor and sensory cortex, which could participate in potential effects of microstimulation in S1 cortex on BMI movements.
2. Long-lasting increase in the efficiency of synaptic transmission in the central nervous system is thought to be a basis of learning and memory. Therefore, there is a likelihood that sensory cortex input participates in motor learning and retention of the learned motor skills.⁹¹
3. There are massive feedback projections from the cortex to the thalamus that modulate sensory information transmission in many ways. Cortical microstimulation may therefore indirectly potentiate or depress tactile responses (via its effect on thalamic receptive fields).⁹²

It is thus necessary to investigate the space and time relationships of neuronal populations in response to various stimulation inputs. Special knowledge and care are required when electrical current is applied to the somatosensory system, which is humans' main mode of interacting with the outside world.

SSNP is based on known functional anatomic connections and physiological interactions between peripheral organs and the brain. One such example is the discriminative sensory pathway for light touch of the hand skin. Next we present the foundations for common SSNP approaches and a brief review of the anatomy of touch sensation.

A. Identifying Discriminative Somatosensory Pathways

Considering the similarity in brains between mammals, the SSNP research aims at external inputs that resemble the natural signals. It hopes to do so by applying electrical current in the discriminative somatosensory pathways, provided the target structure is intact in the given individual.

In short, the somatosensory system, as it relates to common current neuroprosthetic approaches, is mainly the one that focuses the sense of touch conveyed through mechanoreceptors. This special system involves mainly receptors on the glabrous

and hairy skin, which respond to mechanical pressure and distortion.

Mechanoreceptors include Pacinian corpuscles, Meissner corpuscles, Merkel discs, and Ruffini corpuscles. Quickly adapting (QA) Meissner corpuscles are the foremost receptors involved in the detection of vibrotactile stimuli.

The sense of touch arrives at the primary somatosensory cortex via a 3-neuron network that is made of thick, myelinated (fast) neurons. The first neuron (receptor) whose cell body is located in the spinal nerve (dorsal root ganglion) contacts the second one, which has its cell body in the spinal cord or the brain stem, with its axon terminating in the VPL. The third conveys the information from the relay center (VPL) to the S1.

In instances where there is a disruption of the discriminative sensory pathways, as in the case of spinal cord injury or loss of limb, the sense of touch can be potentially replaced by sending artificial signals directly to the brain. That is how BMI devices are being designed to replace lost motor function. It is hoped that such a somatosensory feedback input will improve the performance of BMI devices. Accordingly, microstimulation can be applied in the dorsal column nuclei (DCN), the medulla, the pons, the VPL, or directly in the S1.

Among the many reasons why research in our laboratory has focused on the thalamus and S1, the most cogent include the following:

1. Most of the patients who can potentially benefit from SSNP have spinal cord or lower-structure injuries (leaving the brain intact).
2. There is loss of selectivity (specificity to digits and receptive fields) in lower structures.
3. The cortex is easy to access.
4. Thalamic implants are often common places for deep-brain stimulation. Hence, existing technological and medical knowledge can potentially jump-start SSNP research.

B. Cortical Microstimulation

It is intuitive that guidelines about optimal limits and expectations about the resulting neural response

need to be followed by SSNP approaches. However, these efforts ultimately focus on whatever stimulation results in good behavioral response and an effective BMI feedback.

Psychophysical experiments have contrasted the capacities of humans and monkeys to discriminate the frequencies of mechanical sinusoids delivered to the glabrous skin of the hand. Accordingly, some experiments have shown that monkeys can discriminate between trains of different frequencies and detect brief stimulus trains at relatively low currents.⁸⁰ These results have led some to conclude that behavioral responses (psychophysically quantified during discrimination) produced by natural and artificial stimuli are indistinguishable, at least when one considers the successful manipulation of the neural code underlying the sensation of flutter.⁷⁶

Human subjects and monkeys have been described as having similar abilities to make discriminations on passive or active stimuli. Such capacity has been attributed to QA S1 neurons. The electrical activities of QA and slowly adapting (SA) neurons of postcentral areas 3b and 1 (monkey brain) have been recorded during stimuli-guided behavioral tasks (discriminations). It has been shown that, while there is a resemblance between the two (SA and QA), the SA system plays no role in frequency discrimination in the sense of flutter.⁹³

QA Meissner corpuscles area 3b S1 neurons are strongly entrained by periodic flutter. There is also a match between the input frequency and the firing probability that oscillates at the given frequency, which could be the basis of discrimination between high- and low-frequency flutter stimuli.⁷⁶ These neurons have evoked spike trains, typically with a periodic structure and with highly regular time differences between spikes.⁸⁴ A few laboratories are involved in this research focus, including investigating how attention⁹⁴ or ongoing oscillations^{95–97} can modulate such responses.

C. Ventral Posterolateral Thalamus

Complex actions of movement such as reaching and manipulating objects make use of proprioceptive, cutaneous, and visual feedback. All animals need a

way of providing such information to the central nervous system (CNS). Primates have a visual feedback, whereas the eye position in rodents make visually receiving limb positions difficult. Therefore, rodents have to use proprioception and other senses when moving their limbs and manipulating their food.^{98,99}

Proprioception is essential in the 2 common research subjects: rats and monkeys. Yet there is not enough information about the organization and sensory responses of the VPL, which is the somatosensory relay center. One reason for the lack of data is that, although monkey and rat VPL have similar subdivisions, there are slight differences (monkey) and some rotations (rat) between species. Such inconsistency makes generalizations and knowledge transfer between species rather difficult. Our laboratory at the Downstate Medical Center of the State University of New York,⁷⁸ however, has used electrophysiological techniques to map neural responses to both cutaneous and proprioceptive peripheral stimulation in the rat VPL. We have called for investigations to determine whether its putative subnuclei (rostral VPL [rVPL], middle VPL [mVPL], and caudal VPL [cVPL]) are in fact separable and distinct.⁷⁸

D. Monkey and Rat VPL Have Similar Subdivisions But Slight Differences and Some Rotations

A few electrophysiological^{100–102} and thalamocortical anatomic studies¹⁰³ of the rat VPL have suggested delineations of cutaneous representations. The C7 rat spinal cord lamina I and lamina II are thought to project to the caudal portion of the VPL.¹⁰⁴ Accordingly, 2 pathways arising from the spinal cord have been described. They both go to the ventroposterior nucleus (VP), comprising the ventral posteromedial nucleus and the VPL. One goes to the caudal portion and the other to the rostral VP,¹⁰⁵ the region receiving most of its projections from the DCN. With regard to response, the caudal VP (designated *s2*) has been reported as having responses that are less focal and bilateral.¹⁰² By comparison, the rostral portion (designated *s1*) has contralateral focal receptive fields.

These differences may imply a modular and subnucleated VP. Research using the rat model has

confirmed the presence of a proprioceptive thalamic nucleus in the rat rVPL (–2- to –2.5-mm bregma).⁷⁸ This finding indicates that it may be the evolutionary predecessor of the primate superior ventroposterior thalamus. It also suggested that the mVPL (–2.5- to –3.2-mm bregma) may be the equivalent of the primate VPL where the somatosensory receptive fields are focal. The cVPL (–3.2- to –4.0-mm bregma) is reported as having broad receptive fields and receiving dense restricted projections from lamina I. It is described as possibly being the evolutionary predecessor of the ventroposterior inferior nucleus and the posterior part of the ventral medial nucleus.

Multiple inputs to each of the subdivisions of the S1 in halothane-anesthetized cats have been shown to be carried by populations of neurons located in the ventrobasal thalamus.¹⁰⁶ Another study by Shin and Chapin¹⁰⁷ focused on thalamic somatosensory relay nuclei in the rat medial and lateral ventroposterior (ventral posteromedial nucleus and VPL), ventrolateral (VL), posterior, and reticular thalamic nuclei. They investigated the spatial distribution of the S1 cortical influences and concluded that the spatial pattern of corticothalamic influences seems to reciprocate the known thalamocortical connection patterns.

As one of the candidates for microstimulation for neuroprosthetic feedback, definitive answers about the characteristics of VPL response and its effects in other areas are needed. Studying cortical and VPL responses to microstimulation may help to segregate connections and functional mappings. We discuss below the need for and essential methodologies of carrying out such research.

E. Thalamic Microstimulation

The ultimate research goal in BMI laboratories is to create a prosthetic device that allows the subject to create a smooth movement profile. Such a task will be greatly enhanced by sensory feedback. It has been suggested that thalamic units related to arm movements are confined to the VL thalamus arm area, indicating its role in the initiation of activity in muscles controlling body posture and discrete limb movements.¹⁰⁸ However, few data elucidating the

movement-related roles of the VPL and its subunits are available in the literature. Research is limited despite the fact that rat data have shown⁷⁸ the VPL to be involved in proprioceptive and cutaneous feedback, 2 important components of reaching and grasping.

The amount of literature available regarding thalamic microstimulation for neural prosthesis is rather sparse, which could be because of several possible reasons. One concern could be that thalamic microstimulation has been shown to evoke phantom limb sensations in amputees. The sensations are partially described by somatotopically inappropriate projections to the reorganization of cortical somatotopy following digit or paw amputation and nerve sectioning.¹⁰⁹ There are also reports of thalamic microstimulation causing eye movements,¹¹⁰ paresthesias,^{111,112} tingling,¹¹³ a tingling sensation in the leg coinciding with stimulation-associated angina,¹¹⁴ and sensations similar to those of mechanical stimulation and movements (from microstimulation of the somatic sensory ventral caudal [Vc] nucleus).⁷² These observations suggest that sensations are evoked by the stimulation of pathways specific to cutaneous and deep mechanoreceptors.⁷² Microstimulation of the core of the Vc nucleus, the posterior portion of the ventral medial nucleus, and other nuclei posterior and inferior to the Vc nucleus evokes cool, warm, and pain sensations.¹¹⁵ Microstimulation in the region posterior to the Vc nucleus evokes chest pain almost identical to that occurring during panic attacks.¹¹⁶ In the region below the Vc nucleus, microstimulation caused visceral pain, in addition to nonpainful, paresthetic cutaneous sensations.¹¹⁷

These side effects, however, have not proven to be obstacles to the possibility of using thalamic microstimulation for somatosensation. Some evidence suggests that thalamic microstimulation has the potential for visual prosthesis.¹¹⁸ Several forms of tremor, including parkinsonian, essential, and intention tremors, have been controlled by deep-brain stimulation of the VL¹¹⁹ and ventralis intermedius¹²⁰ thalamus. There are also some studies of thalamic sensory relay nucleus stimulation.^{121,122} Some were done for the treatment of peripheral deafferentation pain.¹²³ They showed local cerebral blood flow

changes and higher neuronal activity in this nucleus and the S1 in response to sensory stimulation.¹²⁴

As the main sensory relay center, the thalamus is a vital node for somatosensory research. It has great SSNP potential, but its roles in previously observed side effects are not well described. Our understanding in terms of loss of peripheral organs and the receptors themselves requires further refinement. The sensory roles of the VPL subdivisions and the effects of cortical or thalamic microstimulation on movement and sensory potentiation/depression have to be investigated.

III. EXPERIMENTAL APPROACHES

Per the aims of SSNP approaches, the overall design and execution of experiments can be concisely described: They mainly involve delivering natural (touch of hand skin receptors) and artificial (direct microstimulation of the brain) inputs, both performed in the same model animal (primate or rat). The resulting neural responses, which are commonly recorded as spike and local field potential (LFP) activities of S1 units, then are compared and contrasted (Fig. 1).

In our setup, initial baseline neural activity of S1 and VPL units is recorded. Natural touch receptors in the hand (or leg) then are activated by a mechanical stimulator. Natural touch stimulation refers to delivering a single touch to the rat's or monkey's skin surface. This is achieved by driving mechanical stimulators, such as push-type, solenoid, actuator-driven vibrating probes, at a given amplitude and frequency. The probe delivers a single contact to the subject's skin surface (an area of $\sim 1 \text{ mm}^2$) lasting about a millisecond. This approach has generated neural responses to mechanical touch in specific receptive fields. Imperfections, such as delays in motor activation and touch strength, can vary as a result of slight skin or limb movements. These cannot be quantified from the mean typical neural response pattern. To control for such minor variations, stimulations are repeated 180 times (at 2 Hz for 1.5 minutes), and average responses are analyzed. Delays are measured periodically using electromyograms obtained from skin receptors. Event times are occasionally realigned offline if deemed necessary.

Mechanical stimulation then is followed by direct microstimulation in the aforementioned brain

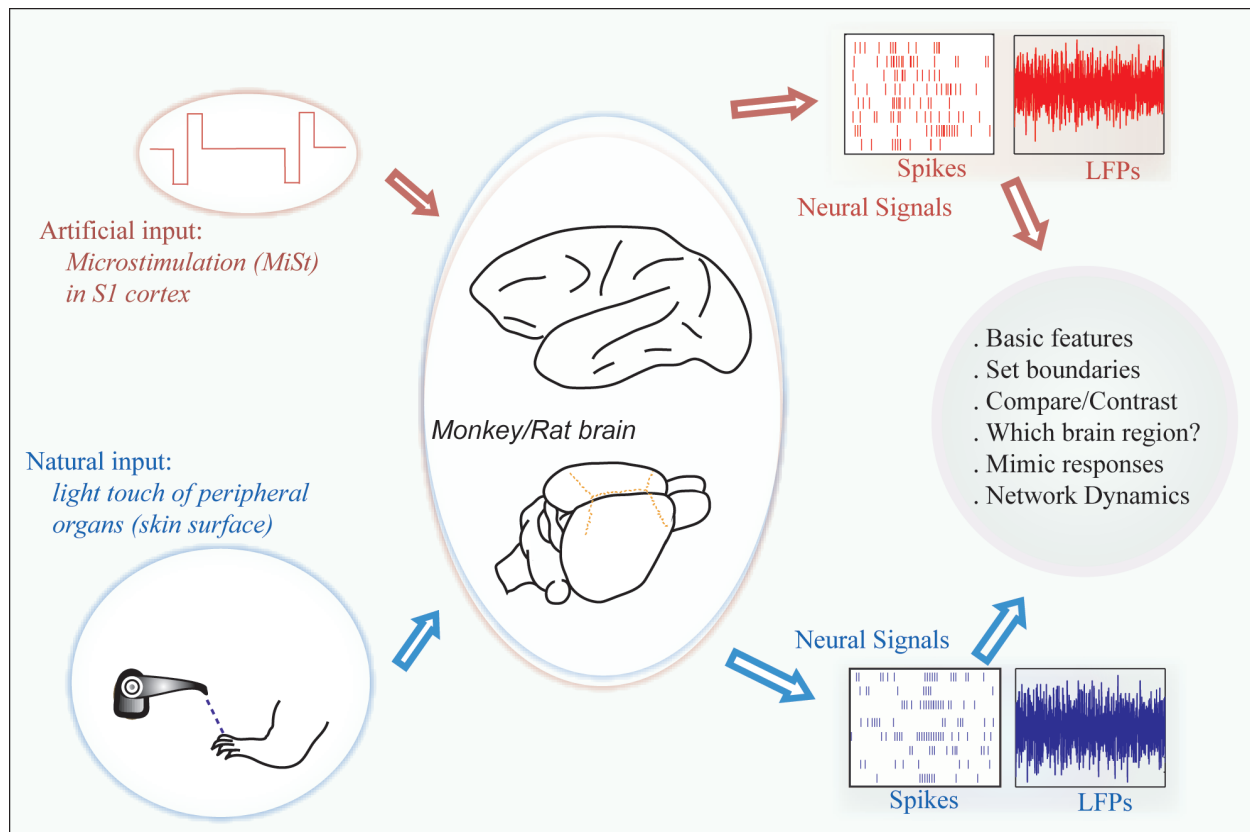


FIG. 1: A general view of experimental approaches to microstimulation for somatosensory neuroprosthesis. LFP, local field potential.

regions. Microstimulation is done using charge-balanced, biphasic and bipolar current pulses (single or double) to mimic natural inputs. While monkeys are always awake, natural stimulation of rats is done under mild anesthesia. It is difficult to keep the hand and fingers still for the tactile input and to prevent the rats from activating touch receptors by laying a paw/finger on objects. Microstimulation is also done in the same condition for a neutral comparison of neural activity. A dose of about 65% of the amount of pentobarbital anesthesia used in surgery (32.5 mg/kg; the surgical dose is 50 mg/kg) was found to be a good compromise dose between maintaining a healthy level of neural activity and perfect stillness of the digits.

The resulting neural data (spikes and LFPs) are recorded and processed offline. Various measures of neural activity then are used to find major differences and similarities and dynamic changes in network interactions. The approach helps to explore

which stimulus parameters to use and which brain region to microstimulate to get the closest match (between natural and artificial) in neural responses. Data analyses, including statistical evaluations, are done using MATLAB (The MathWorks Inc., Natick, MA), Neuroexplorer (NexTechnologies, Littleton, MA), and Offline Sorter (Plexon, Inc., Dallas, TX) software.

A. General Methodologies: Spikes and LFPs

Microelectrode or microwire arrays first are surgically inserted to a depth of a few hundred micrometers in the rat S1, 1 or 1.5 mm in the S1 cortex of monkeys, and around 25 mm (monkey) or 6 mm (rat) to reach the thalamus. Once the animals recover from the surgery and the recordings are stable, basic principles of activation, stimulus parameters, and so on are followed during microstimulation of the cortex

or the thalamus. (See Fig. 1 for a general approach and references 78, 125, and 126 for further details of our surgical and recording methods.)

Despite local limitations and project particulars, there are generally accepted limitations in neural data recording and analysis. In our experiments, neural signals are acquired and amplified in the frequency range of 154–8.8 kHz. The digitization rate is 40 kHz. Spike activity is extracted from the digitized recordings by automated online procedures that identify and sort the spike waveforms into clusters that are presumed to arise from the same unit. These classifications are later refined offline using manual methods. Spike trains are generated by binning the spikes at 1-ms resolutions. LFPs are calculated from extracellular recordings by band-limiting the signal to 3.3–300 Hz. These recordings are done at 2 kHz, which is sufficiently greater than the Nyquist frequency (600 Hz). Various measures are taken to remove stimulation artifacts (these are described in section IV.A., “Stimulus Artifact”).

Rat neural signals are acquired using multiwire arrays from Tucker-Davis Technologies Inc. (TDT; Alachua, FL). In monkeys we use multielectrode arrays from Blackrock Microsystems (Salt Lake City, UT). Both spikes and LFPs are recorded in electrodes that are not used for microstimulation and are not noisy.

B. Firing Rate as a Neural Response Measure

When it comes to measuring neural responses, firing rate modulation (compared with baseline) has been widely accepted and used. Salinas et al.⁸⁴ found that a mechanical vibration task solved by a monkey can be precisely solved by an observer using only the firing rate produced during the stimulation periods.

There is evidence supporting the idea that information from the S1 about the amplitude (rate) and frequency (temporal patterning) of natural vibrations is multiplexed at different time scales by the S1.¹²⁷ Spike timing conveys frequency information for individual afferents and populations, with an additional bonus of afferent class variation in temporal fidelity.¹²⁸

Comparing firing periodicity (the interspike interval) and firing rate as candidate neural codes in vibrotactile tasks showed that periodicity is extremely high in the S1 and the secondary somatosensory cortex, and it is enhanced when it is relevant for behavior. Unlike the firing rate, however, periodicity did not vary with psychophysical performance in single trials, indicating that firing rate is an important component of the neural code for flutter frequency.⁸⁴

On the basis of these ideas, neural data analysis in our experiments is based on firing rate models. These models assume that the average firing response of a neuron to its inputs and the average effect of such firing on the inputs to any other neuron uniquely characterize a neuronal network. These assumptions have simplified computations on the average responses of interest, which are dynamic, but in general they settle to a known steady-state response.^{129,130} Accordingly, the physiological results presented in this article are expressed in terms of firing rate modulation.

C. Selecting Firing Rate Windows for Data Analysis

To obtain consistent measures of similarities and differences in neural responses for different conditions, it is routine to examine responses around a given stimulus, usually called the peristimulus interval. Commonly, a response trace—called the peristimulus time histogram (PSTH)—is made as a spike density function by averaging time-binned activities of neurons over several trials because firing rate variation has proven to be a dependable response detector. It has been used as the main comparison tool for peristimulus neuronal activity modulation. However, these modulations have variable latencies that depend on stimulus conditions. Therefore, specific poststimulus windows are chosen for different recording conditions and analysis methods.

Background (baseline) rates, standard deviations, and other values were always calculated in the same –400- to –15-ms window (before the stimulus). A time point of 400 ms before the stimulus was selected because the longest microstimulation-induced rate change lasted about 100 ms. Because the stimuli

were applied at 2 Hz, 400 ms left enough time after the rate generally returned to and stayed at baseline. Stimulation was delivered with Poisson arrival times. Therefore, upon inspection of the signals, there was no visible ramping up or down of the firing rate caused by input expectation. To remove unknown variations, a 15-ms window before the stimulus was excluded from the analysis of baseline parameters.

Since there was little or no stimulus artifact in the cortical array during natural touch and VPL microstimulation, no window was blacked out. For VPL microstimulation, responses lasted 2–5 ms within the chosen 1- to 11-ms window. For natural stimulation, the 20-ms span immediately after the stimulus was used because that was the average time it took for the longest firing rate modulation to last.

For cortical microstimulation, the maximum artifact width that saturated the preamplifier and blacked-out cortical recording—an interval of 10 ms—preceded the analysis window; that is, a span of 11–21 ms after the stimulus was used for analysis. The cortical microstimulation artifact was not identical for all electrodes. However, such an approach of looking at only the time when there is reliably recorded response modulation in multiple channels is as dependable as considering only the units with no blackout. Moreover, interpolating responses or removing units could cause a loss of information on current spatial spread and inhibition/excitation effects.

IV. BRAIN ELECTRICAL STIMULATION ISSUES: STIMULUS ARTIFACT, INHIBITION, AND OTHER SPATIOTEMPORAL CONSIDERATIONS

A. Stimulus Artifact

A not-yet-fully-solved technical and electrophysiological problem¹³¹ is the difficulty of recording a response during microstimulation. Extracellular action potentials are obscured by the time course of the stimulation artifact.¹³² Depending on the distance of a given electrode from a microstimulation source electrode, the amplification of the input voltage (artifact) overwhelms the recording setup. It makes

recording neural signals (both spikes and LFPs) immediately after the stimulus difficult (Fig. 2) and forces mostly data outside this blackout time to be collected.

Far from the microstimulation electrode, this shutdown lasts only a few milliseconds. It gets larger and larger closer to the stimulus input (Fig. 2). As shown in the insets of Fig. 2 (artifact lengths for different stimulus parameters), an essentially linear relationship between artifact length and pulse width is relatively easily demonstrated, as opposed to artifact length and stimulus amplitude. The incongruity is explained by the fact that very long pulse widths are needed to generate responses at very low amplitudes. The right inset in Fig. 2 also shows that there could be a large artifact even at current amplitudes as low as 1 μ A, given the long pulse duration. These observations do not imply that the artifact is always present or the relationship between electrode location and artifact size is fully understood. As shown in Fig. 2, there is a nonlinear component to the artifact length, making predictions and correction procedures difficult.

Three things are primarily responsible for microstimulation artifacts¹³³: (1) the actual voltage in the tissue at the recording site; (2) the capacitance between the recording and stimulating electrodes; and (3) the AC amplifier overshoot.

Several manufacturers now offer different options and hardware to combat these problems. To our knowledge, no known approach can completely overcome stimulation artifacts. In our laboratory we found that the capacitance is reduced by physically disconnecting recording electrodes at the stimulation site. The AC amplifier overshoot is minimized by staying within a low yet effective range of microstimulation parameters. To reduce artifact, we use stimulation-recording jumpers built in house. They isolate the headstage from the microstimulation current and reduce noise by providing additional grounding. We observed that S1 recordings during cortical microstimulation are affected more by the stimulation artifact than when the thalamus is stimulated. These observations have led us to consider microstimulation of deeper layers of the cortex or lower brain structures. The decrease in noise during thalamic microstimulation could be a result of

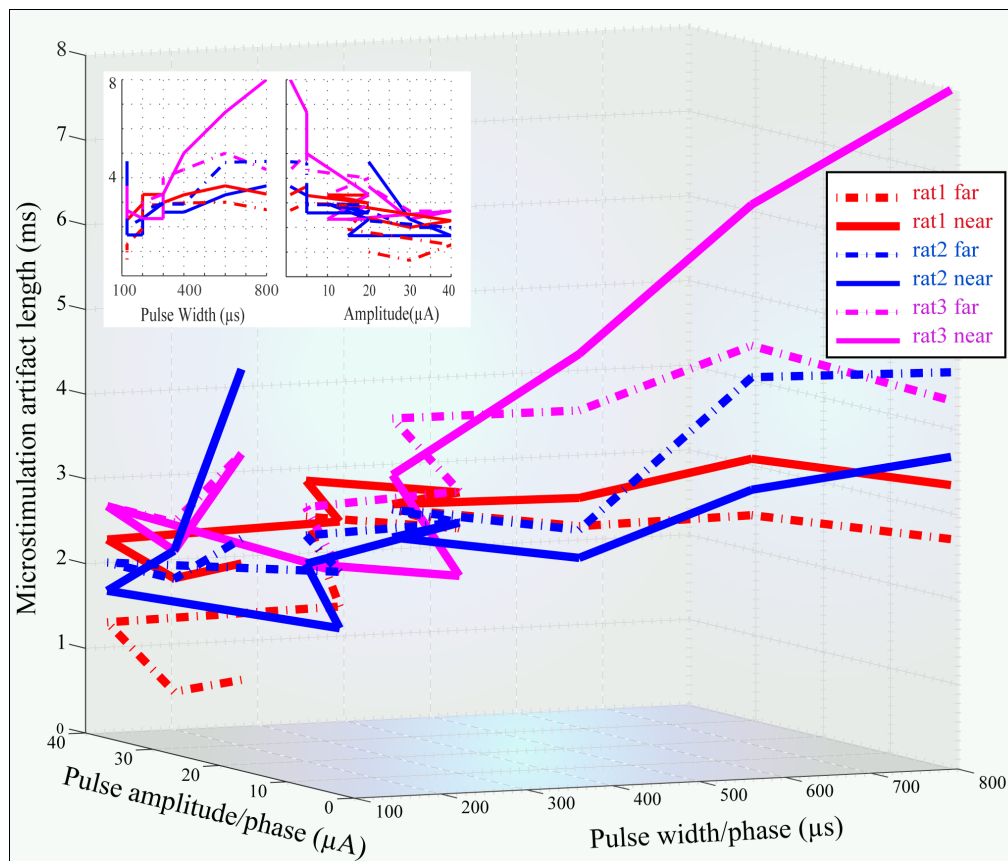


FIG. 2: Microstimulation artifact (in milliseconds; z-axis) in six primary sensory cortex (S1) units (from 3 rats) as a function of stimulus amplitude (in microamperes; y-axis), duration (in microseconds; x-axis), and distance from microstimulation source electrode (dashed and solid lines). All traces are for artifacts recorded during S1 microstimulation. The “near” units (solid lines) were located 0.5 mm from the S1 microstimulation source microwire. The “far” units (dashed lines) were located 1 mm away. Artifact length is defined as the time when the preamplifier is completely saturated and no spike data could be recorded. These session values are calculated from peristimulus time histograms made from counts in the 1-ms time bin, 180 stimulations, delivered at 2 Hz. The insets are side views of the same figure, indicating a more-or-less linear relationship between artifact length and pulse width (left) as opposed to amplitude (right). This is because very long pulse widths are needed to generate responses at very low amplitudes.

(1) the greater separation between the stimulation and recording electrodes; (2) the extent and strength of neural connections and inherent cellular composition, and therefore the amplification and spread of the current; or (3) a combination of these 2 factors.

B. Stimulus Artifact Inhibition and Spatiotemporal Considerations

Effective electrical stimulation and recording resultant activity in highly connected neural substrates is

complicated by the presence of inhibition. Microstimulation activates fibers, causing what some call a spatiotemporal “blur” of activity.¹³⁴

Inhibition during microstimulation has not been fully eliminated because thresholds are difficult to establish and there is a complex dynamic response spectrum. With stimulation of gray matter, as in the case of most cortical stimulation, the exact location of the excitation is not always clear. The usual chronaxie of CNS myelinated fibers is 50–100 μ s, whereas in gray matter it increases to 200–700 μ s.⁵⁷

The difference occurs because the cortex is made of complex and much longer interconnections of local and distant inhibitory and excitatory cells, with strong intracolumn and surface connections.¹³⁵ Excitatory signals are carried to excitatory or inhibitory neurons over distances up to several millimeters via long horizontal fibers present in all layers, but primarily in layers 2, 3, and 5. Although inhibitory neurons have shorter axons (~1 mm),¹³⁶ they can transmit to electrically (and synaptically) interconnected networks of interneurons. Therefore, the electrical stimulus potentially may spread over a large neocortical area.^{137,138} Retrograde, synaptic, or electrical transmission from this highly interconnected network can cause indirect evoked activity, which influences the effect of electrical stimulation of the cortex.¹³⁹

Another issue revolves around an area-specific property of the brain tissue being stimulated; in the case of sensory prosthesis, this is the postcentral gyrus. There is a concern about the motor cortex–like effects of the S1, which calls for investigations of how and why there is a noticeable contrast between the 2 areas. A study by Widener and Cheney¹⁴⁰ found that 93% of the effects on muscle activity were inhibitory when the S1 was stimulated, whereas this was only 25% when the motor cortex was stimulated.

These facts can justify the use of multielectrode techniques, which are better suited to measuring the spatiotemporal resolution needed to evoke functional activity patterns that mimic natural interactions of stimuli applied at different points in space and time.¹⁴¹ However, how a single electrical pulse affects the activity of the surrounding neuronal circuit is not yet understood.

C. Stimulation Parameters

Creating specific behavioral or neural response patterns using microstimulation requires numerous considerations. These include various parameters of the electrical pulse being input to the given brain region and are discussed in this section. In addition to our examples and discussions here, Table 1 provides a summary of the information available in the literature with regard to microstimulation parameters that have been successfully used in diverse conditions and animal species for SSNP.

1. Stimulation Pulse Generation and Main Features

For most neuroscience applications, it has been shown that charge-balanced, constant biphasic and bipolar currents are effective in activating tissue. Enhanced selectivity, with minimized risk of electrode corrosion and tissue injury, has been achieved by biphasic waveforms that are anodic (for fibers) or cathodic (for cells).¹⁵²

Depending on resources, stimulation pulses can be generated by either hardware (such as signal generators) or software using a custom code (such as MATLAB). A stimulation pulse is usually sent as constant voltage input via digital cards (such as the PCI-6711 from National Instruments Corporation, Austin, TX) to an instrument that converts the signals to a constant current and provides opto-electrical isolation (such as the Model 2200 Analog Stimulus Isolator from A-M Systems, Inc., Carlsberg, WA). The choice of the stimulus isolator is critical. At a minimum, a direct current–coupled signal and a stimulus isolator using rechargeable batteries with a built-in recharger are preferred. In our laboratory we use the Model 2200, which has a high maximum bipolar/biphasic output voltage (± 50 V) and a maximum output current of 5 mA, with wide bandwidth and a short rise time (<10 μ s). The instrument can accept a waveform of any shape within its wide bandwidth (direct current to 40 kHz). When delivering computer-generated voltage signals to stimulus isolators, one has to be aware of the output card's voltage range. For instance, the aforementioned NI card has a working voltage limit of ± 10 V. The choice of the parameters and instruments has a considerable effect on microstimulation noise and the integrity of the tissue–electrode interface.

2. Stimulation Current: Amplitude Thresholds and Spatial Considerations

For most purposes, threshold current is defined as the current that evokes an action potential in 50% of trials. Two terms, *chronaxie* and *rheobase*, have been used to describe this concept.¹⁵³ The minimum stimulus strength that produces a response is the rheobase value. Because it is achieved by using the

TABLE 1: Microstimulation Across Various Species, Neural Systems, and Behavioral Paradigms: A Summary

Species	Area	Electrode Used	Pulse Type	Purpose of Stimulation	Pulse Width	Amplitude	Frequency	Duration	Psychophysical Threshold	Source
Monkey	S1 area 3b/1	Utah electrode array	Biphasic pulses, anodic pulse first	Artificial force perception and detection	Phase durations of 200 μ s	5–100 μ A	300 Hz	1 s	10–70 μ A	Berg et al. ¹⁴²
	S1 (3a)	Epoxy-coated tungsten or glass-coated Pt-Ir microelectrodes	Biphasic	Stimuli detection/frequency discrimination	200 μ s	30–50 μ A	15–300 Hz	200–500 ms		London et al. ⁸⁰
Cat	Primary auditory cortex	Utah intracortical electrode array	—	Stimulus detection/lever-press behavior	—	—	—	—	8.5, 8.6, and 11.6 nC/phase	Rousche & Normann ¹⁴³
Rat	S1 (barrel cortex)	Pulled glass-coated platinum tungsten electrodes (80- μ m shank diameter, 23- μ m-diameter metal core; 10- μ m free tip, impedance >1 M Ω)	—	Licking/effects of repetitive stimuli on perception	100–600 μ s	8 μ A (0.8–4.8 nC)	5, 10, 40, and 320 Hz		2.0–2.25 nC	Butovas & Schwarz ¹⁴⁴
Human	Thalamic principal somatic sensory nucleus (ventral caudal; corresponds to monkey ventral posterior)	—	—	To evoke mechanical stimulation and movement sensations	0.2-ms anodal pulse, followed in 0.1 ms by a cathodal pulse	40/50 μ A	300 Hz	1 s	—	Ohara et al. ⁷²
Monkey	Area 3b of S1	1–1.5 M Ω , or 2–4 M Ω ; 80- μ m- and 30- μ m-diameter central cores filled with a tungsten-platinum alloy	Biphasic current pulses, cathodal phase leading	Discriminate the difference in frequency	0.2 ms/phase, 0.05 ms between phases	65–100 μ A			10–30 and 50–200 Hz	Romo et al. ⁷⁶ , Mountcastle et al. ⁹³
Human	Somatosensory and visual cortices	Grids with platinum electrodes; 3-mm diameter; 10 mm apart	—	To produce conscious sensory effects	500 μ s	1–4 mA	60 Hz	500–1000 ms	—	Ray et al. ¹⁴⁵

Monkey	FEF (area 8)	Monopolar stainless steel or tungsten-coated Teflon (0.25 mm thick)	Cathodal-constant, bipolar, anodal, biphasic	Saccade generation	1 ms	0.05–2 mA	200	30-ms pulse train	0.05–2 mA	Robinson & Fuchs ¹⁴⁶
	FEF (area 8)	Glass-coated platinum-iridium electrodes (12- μ m tip, 0.3 M Ω at 1 kHz)	—	Cause saccades under different behavioral states	0.25 ms/phase	5–150 μ A	350 Hz	70 ms		Goldberg et al. ¹⁴⁷
	SC deep layers	Glass-insulated tungsten microelectrodes (2- μ m tip diameter, 0.5–1.5 M Ω or 1- μ m tip diameter, 2–3 M Ω at 100 Hz)	Cathodal pulse	Induce arm movement	400 μ s	10–50 μ A	200–300 Hz	200–500 ms	Threshold to elicit arm movements: 30 μ A (median)	Philipp & Hoffmann ^{148,7}
	Primary motor/premotor cortex	Varnish-coated tungsten microelectrode (impedance: 0.5–2 M Ω)	Biphasic or cathodal	Cause complex arm movement	Negative followed by a positive phase; each phase was 0.2 ms in duration	25–150 μ A	50 and 400 Hz	500 ms	Muscle twitch on half the trials; primary motor cortex (2–12 μ A), premotor cortex (18–100 μ A)	Graziano et al. ^{149,150}
	Primary motor/premotor cortex (as above)	—	—	Create lesions	—	20 μ A	—	30 s	—	Graziano et al. ¹⁴⁹
	Areas 35, 36	Glass-coated iron-cobalt-nickel alloy (elgiloy) microelectrode (0.3–0.8 M Ω impedance)	Anodic direct currents	Mark recording locations using metal deposition	—	2–10 μ A	—	3–30 minutes	—	Koyano et al. ¹¹⁵¹

FEF, frontal eye field; SC, superior colliculus.

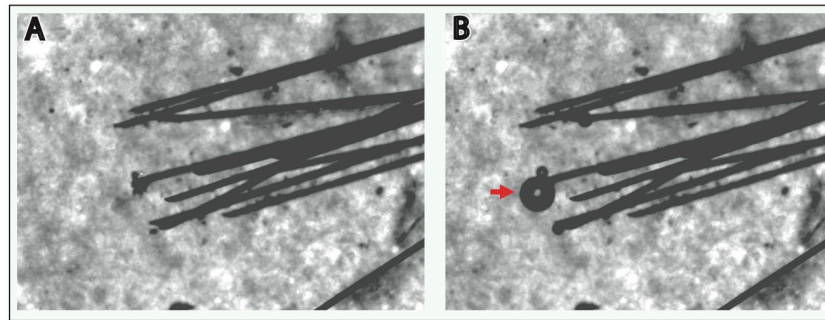


FIG. 3: A light microscopy image (magnification $\times 3.5$) of air bubble formation, an adverse effect of high current microstimulation. Tungsten microwires ($50\ \mu\text{m}$ thick) from Tucker-Davis Technologies, Inc. were immersed in saline (A), and current amplitude was gradually increased until gas bubbles started to appear (B, red arrow). A current amplitude of $750\ \mu\text{A}$ (about 10 times the maximum amount we use for our in vivo experiments, at a $200\text{-}\mu\text{s}$ pulse width and delivered at $100\ \text{Hz}$) was needed before bubbles formed at the tip of the stimulating microwire.

minimum intensity possible, this value is inherently of long duration.

Because the same response can be achieved by adjusting the time variable of the stimulation current, the stimulus duration that causes a response while the stimulus strength is set to exactly double the rheobase is defined as chronaxie. To establish these strength and time constants, one has to investigate the particulars of rheobase-current values (ampere threshold)¹⁵⁴ and chronaxies for the given condition.

The upper current limit for stimulation also has to be established. Damage to electrodes caused by increased current generally precedes damage to tissues. In our laboratory we routinely used parameters (25- to $100\text{-}\mu\text{A}$ amplitude at a $200\text{-}\mu\text{s}$ pulse width) that caused no irreversible damage to either the stimulating or recording electrodes. The damage is usually assumed to be a nonreversible Faradaic reaction. In our laboratory, daily monitored stimulations through the same pair of microelectrodes manufactured by Blackrock Microsystems was continued for 5 consecutive days (about $10\ \text{min/day}$) before a decline in their recording or stimulating capacity was observed. Moreover, microelectrodes/microwires used for microstimulation were designated for stimulation alone, not for recording. All of the work was done using the charge-balancing method, which is believed to be safe. Subsequent histological work showed no visible tissue injury.

In general, only infrequently does stimulation cause adverse effects such as lesions. For example, damage to pyramidal tract neurons occurs only after 10 minutes of continuous microstimulation using an $80\text{-}\mu\text{A}$ amplitude and a $200\text{-}\mu\text{s}$ pulse width delivered at $333\ \text{Hz}$.¹⁵⁵ Data from another study¹⁵⁶ that used charge-balanced biphasic pulse microstimulation ($3000\ \text{nC/mm}^2/\text{phase}$) delivered for 24 hours at $20\ \text{Hz}$ showed the loss of pyramidal tract evoked potential, although there was no visible tissue damage. Another study showed that the threshold for neuronal damage was $300\ \text{nC/phase}$ for charge-balanced biphasic pulses delivered for 36 hours at $50\ \text{Hz}$ through a 1-mm^2 electrode.^{157,158} Our parameters (maximum tested, $20\ \text{nC/phase}$) are well below the range causing tissue damage.

One recommended procedure for setting the boundary parameters is to determine the threshold where oxygen and hydrogen bubbles occur. These accumulate in the brain when too much current is applied.¹⁴³ When this was tested using a rat microwire array in saline, a current amplitude of $750\ \mu\text{A}$ (at a $200\text{-}\mu\text{s}$ pulse width, delivered at $100\ \text{Hz}$) was needed before bubbles formed at the tip of the stimulating microwire (red arrow in Fig. 3B). This amplitude is 7.5 times the maximum current ($100\ \mu\text{A}$) ever used during our in vivo brain stimulation experiments.

Initial parameters were chosen based on the available literature and previous results from mon-

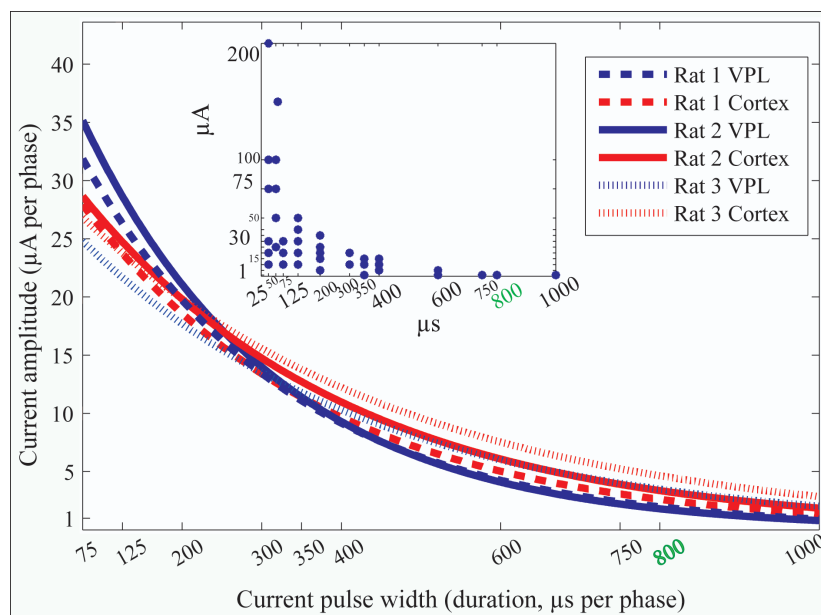


FIG. 4: Current amplitude–duration curves for microstimulation of the primary sensory cortex (cortex) and the ventral posterolateral thalamus (VPL) in 3 rats. A set of amplitude–duration combination points (inset) were tested for evoking above-baseline neural activity, and curves were fit by an exponential model ($f(x) = a \cdot \exp(b \cdot x)$). Activations are as observed in peristimulus time histograms, which were made from counts in a 1-ms time bin, 180 stimulations delivered at 2 Hz.

keys and rats in the laboratory (see section IV.C.2 and Fig. 4). We used microstimulation current ranges that evoked cortical activity resembling that observed during natural stimulations. Most related studies mainly involve behavioral measures to establish effective combinations (see Table 1 for examples). These values are comparable to our typically used parameters (25- to 100- μ A amplitude, 200- μ s pulse width).

The threshold current and its value per neuron is proportional to the square of the distance between the neuron and the tip of the electrode. Depending on the location of a neuron and the electrode tip size, this value can vary. For instance, transsynaptic effects of microstimulation are greater across layers than along layers.⁵⁸ The connectivity of a neuron and the strength of its synapses have a significant influence on the quality of a recording. The following formulas, parameters, and electrode specifications give an overview of the microstimulation environment.

Near the electrode tip, the current density (ID):

$$ID = \frac{I}{\pi r^2} \quad (1)$$

is proportional to distance from the tip and amount of current (I) injected. The surface area (πr^2) of a stimulating electrode has to be known. Smaller tips have smaller radii (r) and thus produce greater current density. Therefore, they recruit more neurons. Consequently, they are commonly used for focused activation of brain tissue, and therefore the term *microstimulation* is used.

Electrode tips are now always flat (blunt cut), however, and surface area calculations have to take the actual shape into consideration. Typical electrodes have angled cuts (i.e., sharpened tips) so they can be inserted easily and cleanly into a given tissue. As a result, their recording behavior and the microstimulating current density near their tips change. In such instances, effective surface area (ESA) is calculated based on the electrode diameter ($\emptyset$) and the angle of the cut (Θ) using the following equation:

$$ESA = \pi \frac{\phi^2}{\cos(\theta)} \quad (2)$$

Accordingly, the rat microwires we used had surface areas of 1711 or 3927 μm^2 (33- and 50- μm -thick wires, respectively, both cut at 60°). The Blackrock Microsystems microelectrodes (for monkey implants) have an active surface area of 2000–3000 μm^2 . These calculations change when there is a special coating on the active (effective) surface area. Coatings of sputtered iridium oxide film and activated iridium oxide film are incorporated to increase the surface area and improve recording and/or microstimulation capacities.

Because it was not within the scope of our work reported here, the effective distance and type and number of neurons could not be accurately calculated and compared with previously published results. Moreover, microstimulation artifact made it difficult to record activity, especially in the electrodes located near the current source. The neural activity data are, however, in general agreement with the few comparable studies, as discussed below.

As reported by others,¹⁵⁹ current amplitude <50 μA (at a 200- μs pulse width, roughly equivalent to 10 nC) fails to activate cells within the cortex (especially) unless the electrode tip is located near the initial axonal segment. Such results were common when microstimulating the S1 using the Blackrock array in monkeys. The current amplitude is as low as 25 μA in monkey VPL microstimulation. Such low current settings were also effective in obtaining neural responses in the rat VPL and S1, most probably because of the low impedance of the TDT microwires used. Measured at 1 kHz, the S1 array has had a baseline impedance with a mean of 25.5 ± 1.47 $\text{G}\Omega/\text{cm}^2$, compared with only 0.87 ± 0.04 $\text{G}\Omega/\text{cm}^2$ of the TDT microwire array.¹⁶⁰

Based on these principles, appropriate stimulus strength–duration parameters were established, as shown in Fig. 4. A single-pulse current intensity of 5–10 nC/phase (25–50 μA at a 200- μs pulse width) was the threshold range when observing cortical response well above baseline and was comparable to activity recorded during the presentation of natural stimuli. A wide range of activation patterns (with regard to arrays) could be observed (Fig. 4) for

current amplitudes as low as 1 μA . Microstimulation artifacts (which were not uniformly linear in the stimulus space) and refractory period were taken into consideration when determining these parameters.

3. Stimulation Current for Lesioning and Marking Recoding Sites

Microstimulation also is used for purposes other than neural excitation for SSNP or controlling body functions. Sometimes there is a need to actively cause a brain lesion to mark a specific brain area, either to mark recording sites or completely damage it for other research purposes. We make electrolytic lesions by passing 15 μA of current using bipolar stimulation with a biphasic square-wave input (20 s for each phase). After creating the lesion, magnetic resonance imaging (MRI)/computed tomography scans can be used to visualize and assess the damage. In instances where the brain is collected from the subject after euthanasia and proper fixation, Nissl-stained images of brain sections through the desired structures (such as the VPL from our experimental monkeys) with an electrolytic lesion are analyzed (Fig. 5A).

Another method for marking locations in the brain is the practice of depositing metals using elgiloy electrodes and passing current. This method has the advantage of not causing lesions but requires the use of MRI machines, which is not always practical because some subjects could be implanted with materials that are not compatible with magnetic resonance. In short, this approach involves creating MRI-detectable metal deposit marks by electrolysis on an elgiloy microelectrode tip, as shown in Fig. 5B (passing a 5- μA current for 5 min through 75- μm -thick electrodes from MicroProbes for Life Science, Inc., Gaithersburg, MD). Table 1 includes research on making lesions from other laboratories.

V. MAJOR DIFFERENCES AND SIMILARITIES BETWEEN NATURAL, CORTICAL, AND VPL ELECTRICAL STIMULATIONS

Most of the results discussed in this review focus on using microstimulation to activate different neuronal populations and for comparing natural

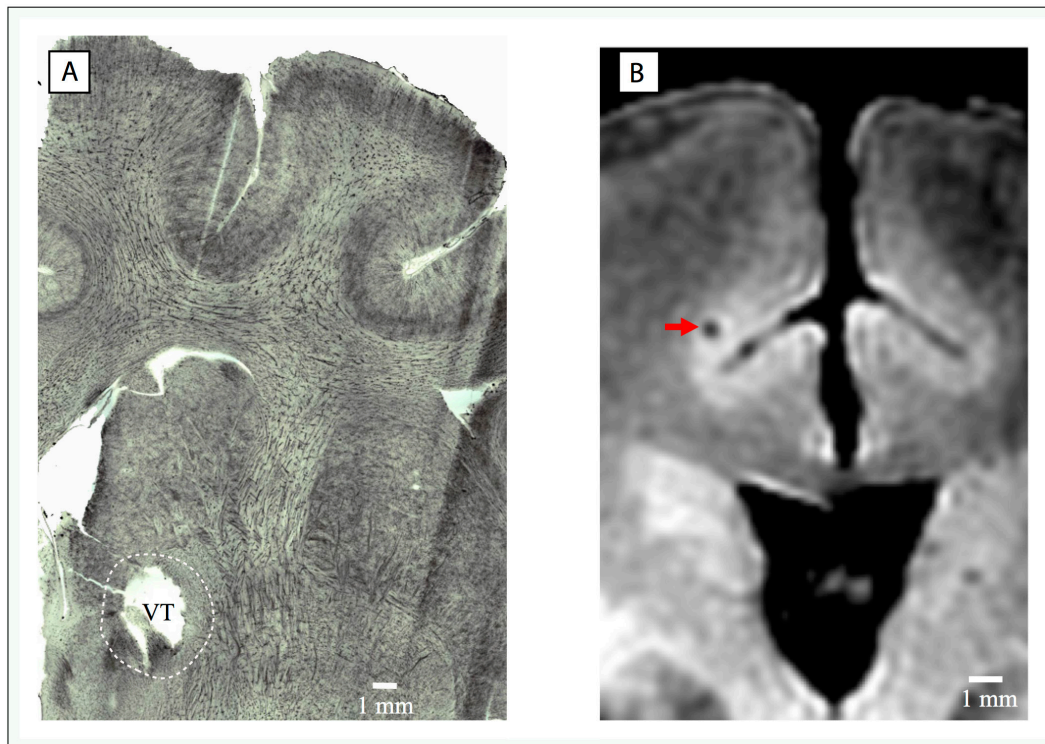


FIG. 5: A lesion and metal deposition caused by microstimulation. A: A Nissl-stained image of a coronal section through fixed brain of an experimental monkey with an electrolytic lesion made on the ventral thalamus (VT). B: A coronal magnetic resonance (T_1 -weighted) image of fixed monkey brain with an elgiloy deposit mark (red arrow). The mark was made by passing a 5- μ A anodic current through 75- μ m-thick elgiloy electrodes (MicroProbes for Life Science, Inc., Gaithersburg, MD) for 5 minutes. Lesions were made at Downstate Medical Center and metal deposition was done at Columbia University. Magnetic resonance images (MRIs) were taken at the New York State Psychiatric Institute MRI center on a GE Medical Systems Discovery MR750 3-T scanner.

and artificial somatosensory inputs. “Bias” ensued because, as discussed in the Introduction, there is a need for BMI interventions. Some of the research in this area carried out at Downstate Medical Center is currently aimed at (1) finding differences in the response from and better candidates for stimulating in the brain for sensory feedback via BMIs, the S1, or the VPL; and (2) examining how the dynamics of causal interactions between S1 and VPL networks change in response to varying inputs.

There are at least 3 major differences between cortical responses evoked by natural touch and microstimulation (Fig. 6). First, the neural response to cortical microstimulation as measured by PSTHs shows an inhibition (several channels) preceded by a

sharp increase in firing rate (in some channels). Cortical inhibition is generally absent during VPL microstimulation. During cortical microstimulation, there is only a short disruption in the recording of VPL activity, which is caused by amplifier saturation by stimulus artifact, but there is no visible excitation of VPL units (data not shown). Second, the high peak, which is common for both cortical and VPL microstimulations, appears in a very narrow window (typically 2–5 ms, compared with 5–10 ms caused by natural touch). Third, natural touch shows sustained response compared to microstimulation (Fig. 6, second row).

There is in effect little or no cortical inhibition during VPL microstimulation (Fig. 6, rows 3–7) at



FIG. 6: Peristimulus time histograms (PSTHs) of single cortical units (3 monkey primary sensory cortex units are shown) under different conditions, showing the main differences and similarities between baseline (first row), natural stimulation (second row), and artificial stimulation. PSTHs show averages of neuronal firing rates in repeated trials, aligned at a stimulus/event. Superimposing repeated trials (presentations) and showing their means helps to explain average responses with respect to what was presented. PSTHs also are called perievent time histograms. In this figure, the dark areas represent neural unit responses. The y-axis shows the firing rate (spikes per second). The particulars of the stimulation condition are shown in the first column. These PSTHs were calculated from counts in a 1-ms time bin (180 stimulations) with stimulation delivered at time 0 (x-axis, red dashed, vertical line). The first unit (unit 40, column A) was selected because it was excited by most conditions, as opposed to some cortical units that were inhibited by cortical microstimulation (column B) or typical ones that were both excited and inhibited depending on pulse parameters (column C). Columns B and C are also units that responded to only a few stimulation conditions. All stimulations were delivered at 2 Hz, and double pulses had a 600-Hz intraburst frequency. Note that the x-axis (time) is expanded in column A. Blackouts caused by inhibition and stimulus artifact (red arrows) are seen with primary sensory cortex microstimulation conditions (bottom 5 rows, especially in units 26 and 60, columns B and C). Natural touch responses are relatively wider (second row). Pulse width (PW) is shown when it changed from 200 μ s. VPL, ventral posterolateral thalamus.

these stimulus parameters. At higher amplitudes, such as 100 μA (a 200- μs pulse width at 2 Hz; data shown in subsequent figures), there is an initial excitation followed by a marked depression that lasts about 100 ms.

Extending these observations to other stimulation conditions, it becomes apparent that during cortical microstimulation there is a clear inhibition time in several cortical units as well as an increase in some units (a period of silent/reduced activity after time 0 in Fig. 6, bottom, column A). By taking the example of a single cortical unit that responded to most of the stimulation conditions (Fig. 6, unit 40, column A) and 2 that did not (Fig. 6, units 26 and 60, columns B and C), the differences and similarities for all of the conditions can be seen.

Figure 7 shows rasters and PSTHs of neural spiking for all cortical units during the various stimulation conditions. Units are arranged (y -axis) by their ranking of when the peak firing rate appeared in the short poststimulus time window. Units where the peaks appear earlier are plotted in the top rows.

Although the unit rankings based on peak time differ among the conditions, the overall population trend stays the same. Note the differences in y -axis labels (Fig. 7), which show the unit number for a selected few and the semidiagonal wave of peak responses after the event (running top middle to bottom right).

Cortical microstimulation results, on average, in an inhibitory response that is more or less proportional to the current amplitude (Fig. 6, rows 8–12). The inhibitory effect seems to last much longer than 30 ms, which is also the average time it takes to settle back to baseline after natural touch stimulation. At high amplitudes (100 μA), thalamic microstimulation results in a long-lasting (~100 ms) inhibition in cortical responses (Fig. 7, row 2, column 3). The silence (inhibition) is preceded by a sharp (~10 ms) excitatory response. During the interstimulus interval, inhibitions are sometimes followed by another wave of “mini-excitation.” It happens about 100 ms later and lasts around 40 ms. The mini-excitation in turn is followed by its own “mini-depression” (~30 ms). Note that the peak z score is higher for the maximum current microstimulation (100 μA) condition (Fig. 8).

Similarities mainly include a poststimulus increase in the firing rate of some units in both cases, whereas certain units in the population do not react at all. There is a marked period of inhibition during cortical microstimulation (a silent period after the event) of several cortical units. In both natural and microstimulation conditions, an analogous pattern of ensemble activity begins to emerge, which is true at least when considering only the modulation of firing rate with respect to event time. From these results, and assuming unambiguous representations are based on population codes,¹⁶¹ microstimulation seems to have the potential to resemble natural inputs.

It has to be noted, however, that the rate depression observed during VPL microstimulation, although lasting longer, is not as deep as that occurring during cortical microstimulation. Another caveat here is that cortical microstimulation depression (dips in the red traces in Fig. 7) is also caused by a stimulus artifact that prevents recording of the responses during the immediate poststimulus window.

Another common occurrence was that modulation is not uniform over baseline firing rate rankings, indicating that the inherent firing rate may play a role. Inherent firing rate differences, especially in connection with very low-frequency and very high-frequency oscillations, should be investigated further.

One advantage of microelectrode arrays is that electrode spacing is not affected by surgical procedures, making spatial analysis possible. The Blackrock cortical array thus was used to study how activity patterns change along a given direction. The analysis was accomplished by taking a row of 10 electrodes, identifying the largest peak normalized (z score) within the row, and observing how these values change going out to the periphery of the array. Figure 8 shows the spread of the peak z score values for 2 perpendicular rows from the array in the anteroposterior (Fig. 8A; anterior is toward the central sulcus) and mediolateral (Fig. 8B) axes. In almost all the conditions the spread and falling off of activity maintains a comparable shape. The peak activity for a given row is localized to just about 2 electrodes covering an area of approximately 0.8 mm. Significantly, it tapers off close to zero in both directions and along both axes.

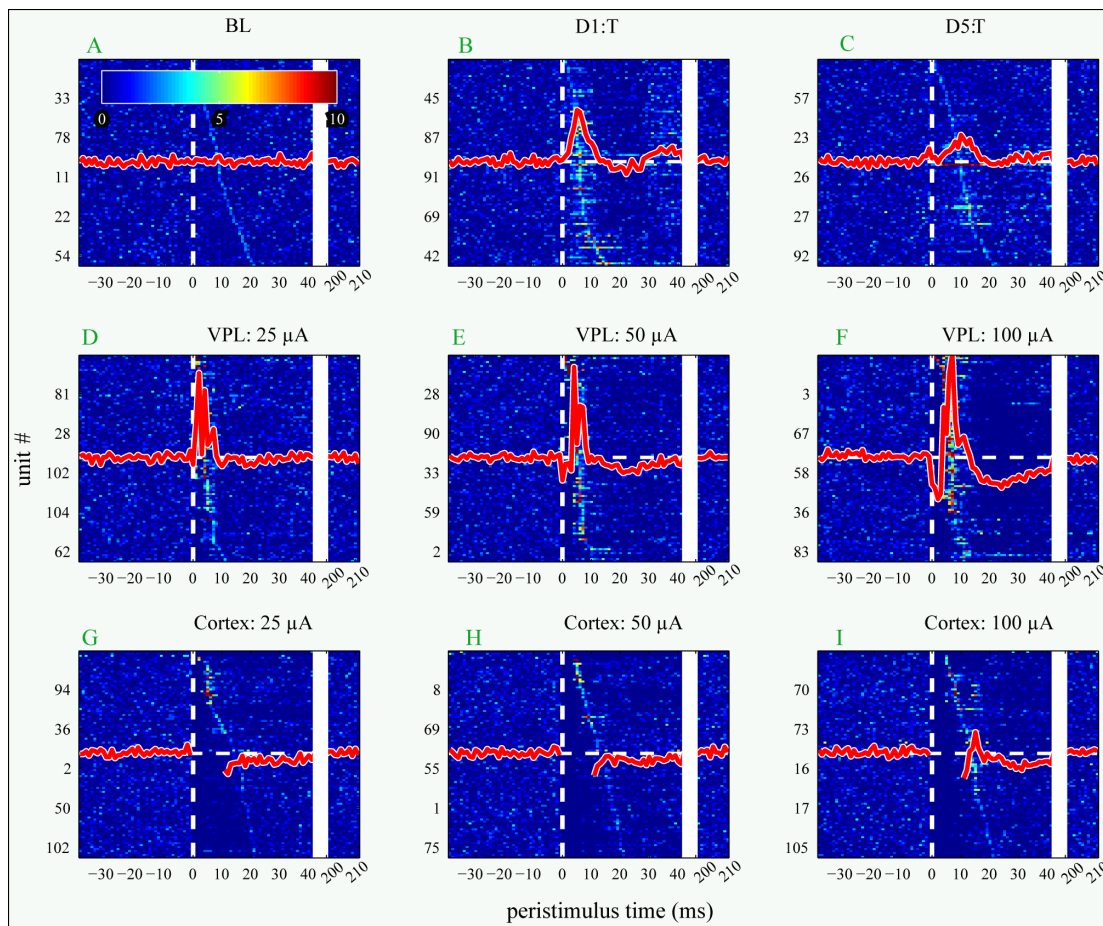


FIG. 7: Population peristimulus time histograms (PSTHs) for cortical (monkey primary sensory cortex) units (y-axis) calculated from counts in the 1-ms time bin (180 stimulations). Stimulation was delivered at time 0 (x-axis, white dashed, vertical lines). Each panel is a population response to a condition. Each row is the normalized firing rate for a given unit. The white dashed, horizontal line is zero for the sum of instantaneous firing rates of all units (red trace). The breaks in the red traces in panels G–I (cortical microstimulation) depict the poststimulus time window excluded from analysis because of the blackout of cortical recording caused by stimulus artifact in several cortical channels. Cortical microstimulation causes inhibition (G–I), whereas VPL microstimulation (D–F) causes a natural touch-like population response (B and C, D1 and D5 pads). The condition or pulse amplitude is the title of the panels. All panels are in the same scale, as indicated by the color bar, which shows intensity of firing after normalizing each unit to its own prestimulus firing rate. In the y-axis the units are arranged by their ranking of when the peak firing rate occurred in a short poststimulus time window. This window is the time for natural touch (1–21 ms) and VPL microstimulation (1–11 ms) immediately after the stimulus, and the time after a 10-ms blackout period (taking the extreme case of cortical stimulation artifact) for cortical microstimulation (11–21 ms). As a result, displayed in the top rows are units whose peak response happened earlier. Although the rankings of unit peak time differ among the conditions, the overall population trend stays the same. The trend is shown by the semidiagonal wave of hot colors after the event, running top to bottom. Also, note that at high amplitudes (clearly seen at 100 μ A), thalamic microstimulation causes long-lasting ($\sim$ 100 ms) inhibition in cortical responses. On average, it takes roughly 100 ms to get back to baseline activity after microstimulation (white vertical break). BL: Baseline; T: natural touch of hand digits (D1 and D5). All microstimulations were single pulses with 200- μ s pulse widths and delivered at 2 Hz.

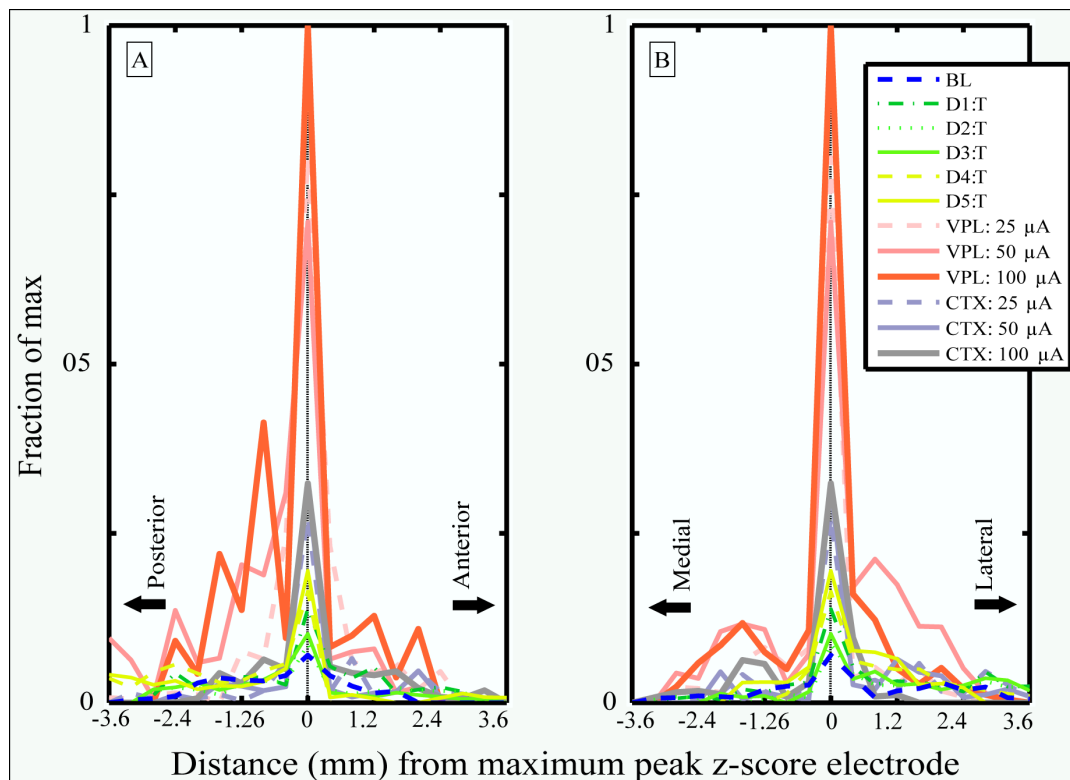


FIG. 8: Peak z scores (y-axis) as a function of distance (millimeters; x-axis) from maximum z score cortical electrode looking in 2 perpendicular directions in a given 10-electrode row (from a Blackrock multi-electrode array implanted in a monkey primary sensory cortex). **A:** Anteroposterior axis (anterior is toward the central sulcus, right of “0” on the x-axis). **B:** Mediolateral axis (medial is left of “0” on the x-axis). These values are calculated from counts in a 1-ms time bin (180 stimulations at 2 Hz). More electrical stimulation parameters are in the legend. BL, baseline; CTX, cortex; T, natural touch of hand digits (D1 to D5); VPL, ventral posterolateral thalamus.

One major concern in brain microstimulation is how it affects underlying/lasting single/population neuronal activity. Thus, our data were tested to see whether microstimulation causes selective changes in the proportion of units that are firing within a given range (rate clusters). Using the Blackrock array in the monkey S1 (hand area) showed that spiking activity is primarily confined to the low-frequency range (15–35 Hz) (Fig. 9), which is in general agreement with previous research.¹⁶² Such a trend was observed over several days and seems to be a stable property of the units. Figure 9 shows that low-frequency units generally are recruited into the response population, irrespective of the recording condition.

A comparison of the mean firing rate within the 30-ms window following stimulation shows no

statistically significant difference between natural touch and VPL microstimulation. However, they both are different from cortical microstimulation (Fig. 10, bottom panel). Multicomparison analyses using balanced one-way analysis of variance indicates 3 pairs of means (baseline, natural and VPL microstimulation, and cortical microstimulation) are significantly different ($P < 0.0005$ for all). Microstimulation in the VPL causes very sharp and high-amplitude neuronal responses (as measured by PSTHs) compared with natural and cortical microstimulation. Tactile input causes a sustained high rate hike after stimulus compared with microstimulation, which commonly gives very sharp responses. Note the small standard errors of the mean (red whiskers in Fig. 10) for the baseline and natural stimulation conditions.

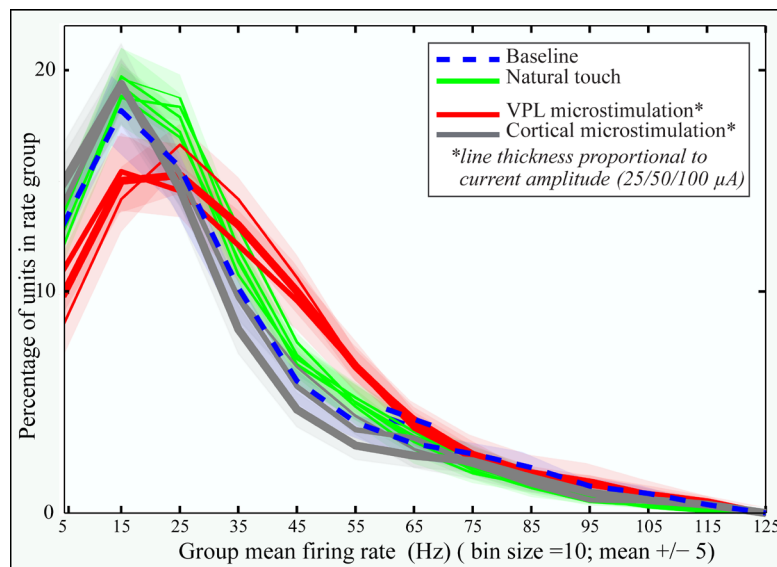


FIG. 9: Change in the percentage of cortical units (y -axis) in a group with a common firing rate range (± 5 group mean on the x -axis) as a function of stimulation condition. Shown are means and standard errors of means from 8 days of the same stimulation and recording condition. The thickness of mean plots for microstimulation conditions is proportional to the current amplitude, as indicated in the legend. VPL, ventral posterolateral thalamus.

Mean firing rates and peak z scores were analyzed using a grand average of all cortical units (88 units) for 4 days of recording and stimulation (180 repetitions $\times$ 4 days = 720 trials per condition). These results hold up even when 8 days of recordings were used for this analysis, despite the fact that 4 days of data were collected when recording and stimulating at different electrode locations (for both cortical and VPL microstimulation). Units included in this analysis were from a semiautomated offline sorting and made use of only those that are common to all recordings (files) for a given day. Some had to be excluded because it is possible for waveform characteristics to drift over the course of the experiments, and consequently some units appear or disappear as the recordings progress.

The overall mean for the entire array is higher for the medium current setting (50 μ A; Fig. 10) because using higher current values causes momentary inhibition in several channels, dampening the grand average. Since not all units respond to a given condition, modulation of the overall firing rate is very restricted (± 1.75 standard deviations from baseline means). Next to baseline, natural stimulation is the least modulated

of all. However, natural stimulation responses have double peaks (approximately 10–12.5 and 37.5–40 ms after stimulus), which could occur because of several reasons. The most obvious one is that the same array has 2 populations of neuronal units. When classified according to the latency of their evoked responses, some (short latency response units) respond before others (long latency response units). On average, short latency response units respond within 5–15 ms and long latency response units within 30–40 ms of a natural (touch) input. Such units have been observed in the forepaw area of the rat S1.^{163,164}

Other important questions are, What is the physiological basis of responses to tactile or artificial inputs? How do such responses relate to ongoing cortical activity? There are now studies showing that sensory responses ride on inherent oscillations that reflect excitability states.⁹⁵ Thus, investigating whether and what kind of microstimulation can be applied to get preferred responses is crucial.

When microstimulation was done in the VPL or S1 using double pulses, clearer responses were recorded in cortical units (compared with single pulses at higher or lower amplitudes; Fig. 6). Espe-

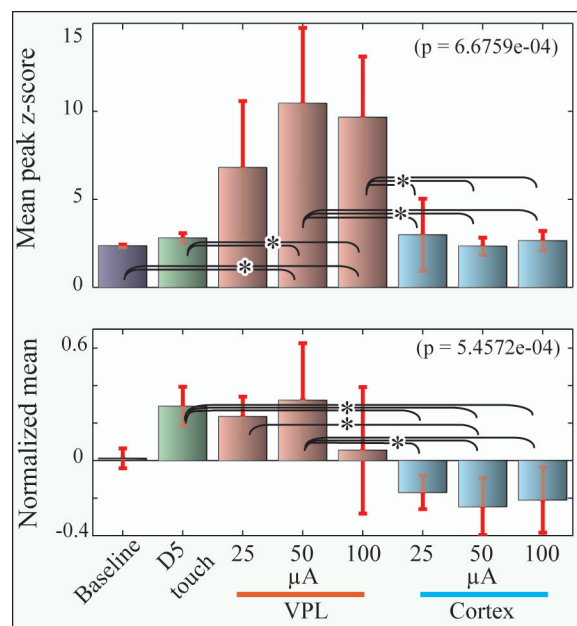


FIG. 10: Multiple comparison test results for cortical response window mean peak z score (top panel) and firing rates (bottom panel) from means of 4 days of independent observations (180 stimulations $\times$ 4 days $\times$ 88 units = 63360 trials). Peak z scores and firing rates of units in poststimulus windows (described in text) at each recording position that are common to all recording conditions were used for this comparison; their standard errors of means are shown as red error bars. The means for each unit are calculated from a self-normalized peristimulus time histogram. A multiple comparison using balanced one-way analysis of variance indicates 3 pairs of means are highly significantly different (connected by horizontal brackets marked with asterisks; $P = 6.6759 \times 10^{-4}$ [top panel] and 5.4572×10^{-4} [bottom panel]). Both microstimulations are single 200- μ s pulses at 2 Hz. Current amplitudes are shown as X-tick mark labels. D5 touch, natural touch of the pad of digit 5. VPL, ventral posterolateral thalamus.

cially effective were double pulses with an intraburst frequency of 600 Hz (4 biphasic pulses per second, members of a pair separated by 1.6 ms). Does this have something to do with the known high-frequency (600 Hz) spike bursts that are observed in monkeys?¹⁶⁵ Or somatosensory evoked human electroencephalographic and magnetoencephalographic responses (also 600 Hz)?^{166,167} Such questions await further investigations.

Thus far we have compared only similarities and differences between natural touch and artificial stimulations of the VPL and S1. However, interactions between and within brain regions can have serious implications in SSNP.

Although it is not possible to sample all neurons involved in a specific behavior, the current ability to record from multiple electrodes enables quantitative measures of neuronal activity to be obtained and provides an opportunity to probe information flow in different networks. Since network interaction is dynamic and dependent on stimulus conditions, experimental paradigms can be implemented to test several hypotheses: (1) functional connectivity measures can be developed as the first tool to model neuronal dynamics; (2) such models can be used to estimate functional interactions; and (3) network organization can be reversed.

To test these hypotheses, LFP signals recorded from the rat S1 and VPL subnuclei (rostral, middle, and caudal) were analyzed for functional interactions. The goal was to find the direction and strength of causal influences within and between these brain regions, as well as how they are differentially affected by specific inputs.

To evaluate such a paradigm, a method that is based on economics and has been gaining popularity in neuroscience, linear Granger causality (LGC),¹⁶⁸ was used. In short, LGC is a statistical concept that tries to explain the relationship between a given signal (x) and another one (y) that is a candidate as a possible cause of the behavior observed in the first (x). Once these linear and one-directional influences are established, interactions are depicted by connecting recording nodes, as shown in Fig. 11; the thickness of wires is proportional to the amount of causal influence. The findings, such as those shown in the wiring architectures in Fig. 11, are strictly functional. Although they may imply structural connectivity, they are not meant to represent anatomic links.

Consequently, the main changes between directional influences under different recording conditions were statistically quantified. To create a grand average to which statistical estimation techniques could be applied, 180 repetitions of a recording/stimulation condition were recorded from 3 rats over 3 days ($n =$

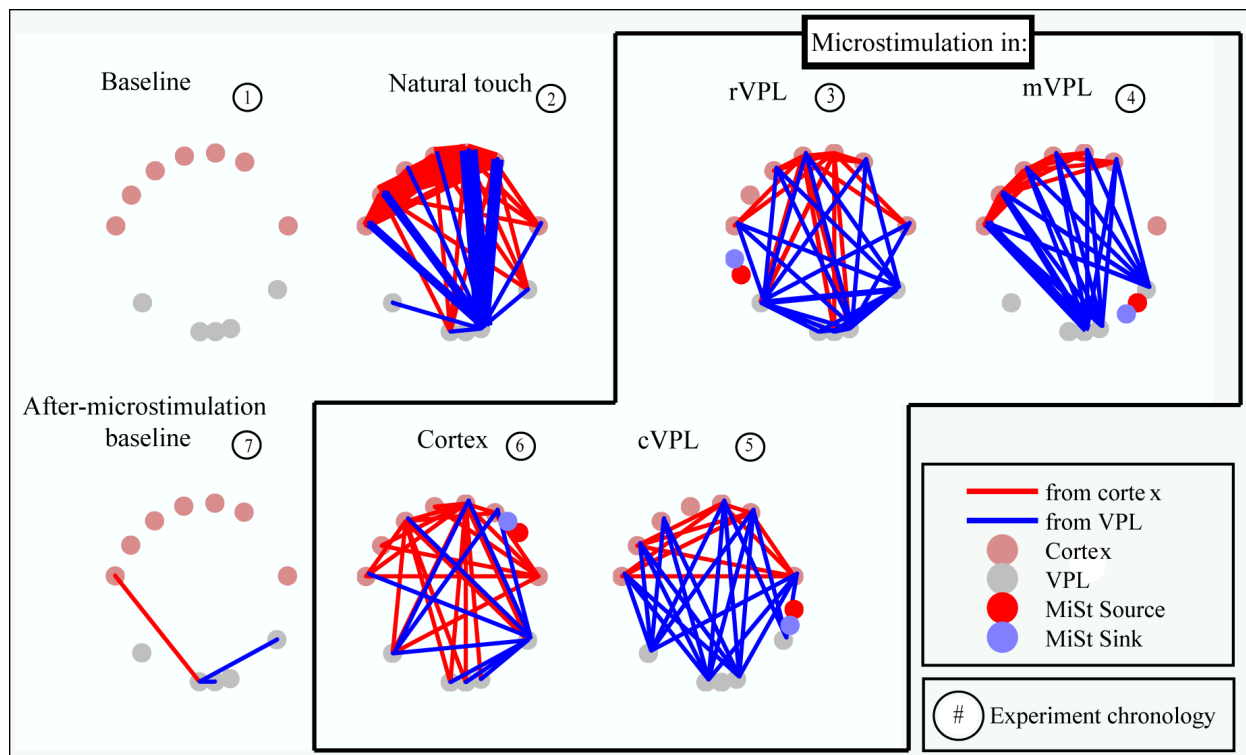


FIG. 11: An example session of significant linear Granger causality (LGC) wirings for different directions of interactions and stimulation conditions. The LGC analysis was done for information flow between a set of 16 cortical (top-tier dots) and 16 ventral posterolateral thalamus (VPL) (bottom) recording sites in an anesthetized rat (data from one session; a mean of 180 trials). Significant transfer of causal information within the set of the recording sites evaluated between each pair of electrodes is shown by connecting the nodes with wires whose thickness is proportional to the strength of the Granger causality. All directions of flow are represented (see the legend). The 3 subnuclei of the VPL—rostral (rVPL), middle (mVPL), and caudal (cVPL)—were microstimulated separately. Electrode positions in the respective brain structures are indicated by each node. Noisy or microstimulation channels were not included in the analysis. Red lines indicate directional influences from cortex to cortex and cortex to VPL; blue lines indicate directional influences from VPL to VPL and VPL to cortex. Panel titles indicate the recording condition. Microstimulation current source and sink are represented by red and blue circles, respectively. All microstimulations occurred at 2 Hz, 25 μ A, and 200 μ s (charge-balanced bipolar and biphasic). MiSt: microstimulation.

$3 \times 3 \times 180 = 1620$ trials). Both within- and inter-subject variabilities were studied using a 2-sample Kolmogorov-Smirnov t test. Nonparametric assessment also was performed using the Kruskal-Wallis and Wilcoxon-Mann-Whitney tests.¹⁶⁹ No statistically significant variations between the subjects were observed ($P > 0.01$). Cortical microstimulation significantly increased cortico-cortical (C-C) interactions. The largest difference was between C-C interactions during cortical microstimulation and baseline conditions ($P < 0.0001$, t test; $P = 0.00005$,

Kolmogorov-Smirnov test). The largest (bulk) causality interaction was between C-C interactions in the cortex, and it was followed by natural touch C-C interactions. However, there was no statistical difference between them. Baseline conditions manifested little or no interaction and, except for a few outliers, had mostly zero means. As a result, cortical microstimulation responses differed from baseline rates both before and after stimulation. Most of the causal communication is contained within the cortex, followed in order of decreasing

probability by corticothalamic, thalamocortical, and thalamo-thalamic influences. When compared with natural stimulation, VPL microstimulation increases thalamocortical information flow. But there is little variability in natural stimulation, and it is different from microstimulation conditions. Moreover, there is little or no statistical difference between baseline causal interactions before and after the stimulus, which implies that, microstimulation and natural touch do not permanently change network activity within 5–10 minutes after the stimulus. It has been shown that microstimulation lasting 30 minutes does not lead to long-term functional effects.¹²⁶

VI. DISCUSSION

Multineuron spikes, and signals obtained through noninvasive methods such as electroencephalography and evoked potentials, have been used for building BMIs. They have been used in human subjects and animal models to control computer cursors and robotic arms, and to generate synthetic speech for individuals with severely impaired motor control and loss of speech.^{155,170} A challenge with most BMIs is that natural movements and motor learning depend on the CNS, which makes use of visual, proprioceptive, and cutaneous information,¹⁷¹ whereas most BMIs depend primarily on natural vision or hearing to guide motor control.¹⁷² When compared with natural motor control, BMI control is perceived to be slow and is not totally predictable. This is caused by either limitations of forward controllers or the lack of proprioceptive and cutaneous feedback.⁸⁰ It is likely that an application of microstimulation can generate suitable somatosensory feedback from the prosthetic devices to the user to alleviate the problem. However, current research has been able to explain neither the mechanisms underlying microstimulation sensations nor the perceptions arising from microstimulation of neural structures.

The majority of sensations induced via microstimulation are described as unnatural and tingling.^{114,173} It is likely that the shortness of cortical responses to microstimulation compared to touch may be the cause of fleeting and tingling sensations. Moreover, factors such as thresholds, conscious

awareness of sensation, and quality of senses are different among individuals. It is thus difficult to compare natural touch with fleeting and tingling sensations.¹⁷³

On the other hand, different tasks use the same somatosensory pathways (cutaneous receptors to the S1) to accomplish a variety of goals, such as a touch sensation and holding an object. This has been demonstrated by what is termed “the wineglass effect,”¹⁷⁴ which is caused when a tiny object slips from a grasping hand. The slips generate vibrations that are detected by cutaneous receptors in the fingertips. The detection creates a cascade of muscle reactions that compensate for the slip and cause a tighter grip. These load compensations, which are faster (latencies of 60–80 ms) than reaction times,¹⁷⁴ seem to be not conscious¹⁷⁵ and have time constants that are different.¹¹² Hence, even though conscious sensation and task performance are not always fused together, the aforementioned behavioral studies indicate the possible use of microstimulation for sending various feedback signals to the brain.

Accordingly, Heming et al.⁷³ recorded neural responses in the VPL and then played these recordings back into this neural region via microstimulation with the hope that such a replay would lead to simulating natural sensations. However, such an approach has not been any more successful than simple high-frequency microstimulation trains. Thus, playing back a previously recorded firing pattern does not guarantee the neural activation that generated it. Moreover, such an activation is made difficult by stimulus artifacts caused by microstimulation. Switching between recording and stimulating at the same point in space is very challenging. Perhaps the use of optogenetic stimulation coupled with traditional neurophysiological recording techniques could overcome some of the obstacles.

The microstimulation strategy presented in this review aims at (1) comparing natural and artificial somatosensory inputs; (2) finding a better candidate brain area for a sensory feedback via BMIs, the S1, or the VPL; and (3) showing how the dynamics of causal interactions between S1 and VPL networks change in response to different inputs. The data presented here and in the literature suggest the following:

There are at least 2 major differences between cortical responses evoked by natural touch and microstimulation. (1) Neural responses to microstimulation (especially cortical) as measured by PSTHs show an overall inhibition highlighted by the peak firing rate. The inhibition and peak responses happen in the window immediately after the stimulus and in comparison to natural touch. (2) The peak firing rate occurs in a very narrow time window immediately after the stimulus (typically 2–5 ms, compared with 5–10 ms in natural touch). Thus, simulating natural responses is not a trivial undertaking. Compared with microstimulation conditions, natural touch causes a very subtle modulation of firing rates (with regard to population).

There are several similarities between natural and artificial stimulation. They include a poststimulus increase in the firing rate of some cortical units following both types of stimulation. However, certain units in the neural population do not react at all. In both conditions a consistently similar rate of firing activity of a neural population begins to emerge, especially in the distribution of the peak firing rate following stimulation. Since unambiguous representations in neural responses are based on population codes,¹⁶¹ the results indicate that microstimulation has the potential to resemble natural inputs.

1. Compared with cortical microstimulation, thalamic microstimulation may have an equal or higher probability of creating cortical responses that are similar to those observed during the presentation of natural inputs. There is a distinct inhibition time common in cortical microstimulation, which is also present during VPL microstimulation at high current amplitudes ($>50 \mu\text{A}$). Compared with natural and cortical stimulation, responses (of cortical units) to VPL microstimulation are markedly sharp and have higher amplitudes (Figs. 6–8). It is possible that thalamic thresholds are not exactly the same as cortical thresholds. More cortical units are activated during VPL microstimulation compared with cortical microstimulation.
2. Stimulus strength and neural response relationships are not linear (Fig. 6). This is especially

true after one effective microstimulation current parameter combination: a 200- μs pulse width and a 50- μA amplitude at 2 Hz. Effective amplitude for VPL microstimulation can be as low as 25 μA .

3. Causal interactions are inherently dynamic and vary with stimulus conditions. In order of probability of occurrence, functional influences are mainly C-C, followed by corticothalamic, thalamocortical, and thalamothalamic in direction¹²⁶ (Fig. 11). Sensitivity to arousal state, DCN gating, and the BMI implications of these modulations need further investigation.

In addition, certain observations that seem to be counterintuitive should be examined carefully. One such observation is that there is a strong functional corticothalamic influence during a few idle (baseline) recording conditions, which could be caused by 4 possible reasons. The first is non-stimulus-related communication between different brain regions (periodic polling), which may be part of an active sensing mechanism.⁹⁷ The presence of such a process can be explored by recording neural signals when subjects are attentively waiting for external inputs. The different conditions (states during these delay periods) may be marked by specific oscillation bands. One has to be careful in using LGC for this purpose because it may give misleading results in the case of causality on a spectral band.¹⁶⁸ The second is that sensory information is modulated by a strong corticothalamic feedback projection.⁹² Third, compared with the thalamus, the cortex has a stronger neuronal interactions,¹⁷⁶ which may potentiate its outgoing influences. Finally, the sampling of natural peripheral inputs by thalamic relay neurons is affected by their driver, the cortex.¹⁷⁷

Stimulation parameters used in microstimulation have to take into consideration inherent cellular properties that may call for the tailoring of individual current inputs. For example, the neural population that is activated during gray-matter stimulation is not well understood.⁵⁷ Nonetheless, some investigators maintain that evoked electrical activity in gray matter is caused by axonal excitation following somata and

dendrite stimulation.¹³⁴ Depending on interconnectivity (“indirect” effects), axonal excitation carrying the activation by ortho- and antidromic conveyance of spikes to areas is thought to mask the direct neuronal activation of somata. A major part of the stimulation is assumed to be carried by local antidromic and/or synaptic effects because there is abundance of axonal elements in the neocortex and most other gray matter of the CNS.¹³⁴ It is important to explore these issues deeper because psychometric performance can be significantly shifted based on the shape of the microstimulation waveforms.¹⁷⁸ Modeling approaches have recently gained some momentum because of the proliferation of multielectrode systems and the need for precise microstimulation.^{179,180}

Further studies should focus on the effects of somatosensory microstimulation on the motor cortex and the VPL. It is likely that the input from the sensory cortex participates in motor learning and retention of the learned motor skills.⁹¹ The efficiency of inhibitory transmission in the thalamocortical neuronal networks show long-term changes in response to microstimulation of the cortex.¹⁸¹ These changes have manifested in the form of long-term potentiation and/or depression of excitatory connections¹⁸² and also occur in the form of intensification or attenuation of the action of a “common source,” a signal that synchronizes the discharges of pairs of neurons.¹⁸²

Long-term potentiation of thalamic inputs to the primary motor cortex (M1) can be induced by coactivation of thalamocortical and C-C afferents,¹⁸³ perhaps because the VL directly synapses on the M1 and there are strong interconnections and anatomic proximity between the S1 and the M1. Accordingly, motor thalamic areas have a role in movement and guidance.¹⁸⁴ It is thus conceivable that somatosensory microstimulation can potentially intrude on movement, especially after extended usage. Synaptic proliferations in the motor cortex have been reported after long-term microstimulation of both the cortex and the thalamus.¹⁸⁵

Microstimulation of other brain areas such as the superior colliculus also has caused arm movements.¹⁴⁸ Microstimulation in the same region (superior colliculus) can cause other effects such as pupil dilation.¹⁸⁶ For somatosensory neuroprosthesis to be applicable in

real-life settings, it ideally has to be free of movement interference and lasting neural alterations.

Sensory information transmission is modulated in many ways by massive feedback projections from the cortex to the thalamus.⁹² Perhaps because of this, cortical microstimulation modulates guinea pig exteroceptive information relay thalamocortical neurons.¹⁸⁷ This modulation may potentiate or depress tactile responses.¹⁸⁸ Further investigation of this possibility is justified, although our data did not show cortical microstimulation causing any visible and long-lasting activity change in the VPL.

Some issues need to be addressed immediately. Artifact-free stimulation and recording setups and strategies have to be designed so that neural data from zero to a few milliseconds immediately following microstimulation inputs is not lost. The artifact blur usually includes what may be the most important response history. Some approaches seem to be promising. Using online adaptive filters to recover neural spikes from stimulation-corrupted records,¹³² or a zero forcing equalizer filter and a patterned pulse stream, have been reported to have the potential of reducing artifact width by up to 70%.¹⁸⁹

Cortical microstimulation has induced inhibition,^{134,190} in contrast to natural touch, which causes cortical activation. In the absence of other targets, new approaches to stimulate S1 with less or no inhibition may prove to be useful. It should not be forgotten, however, that inhibition has been irrelevant for sensory discrimination. Finally, stimulation parameters and spatiotemporal manipulations should be individually tailored for any given area or neural unit, taking into account inherent (baseline) properties.

VII. CONCLUSIONS

Microstimulation has been effectively used in improving our lives for some time. However, using brain microstimulation to generate somatosensory percepts that are similar to those induced via natural inputs is novel and has significant potential in medicine. At a minimum, such an attempt requires an understanding of sensory processing and mapping neural responses to optimized electrical and cutaneous stimulation. Moreover, “similar” can be

highly subjective; natural inputs have a very subtle yet efficient consequence on neural activity, which is understandably difficult to replicate.

Ultimately, effectively sending sensory signals to neuroprosthetic devices is the only grounds for reconciliation. In addition to touch, the potential of external devices with other forms of feedback, including proprioception, 2-dimensional contour and motion (areas 3b and 1), and 3-dimensional shape (areas 2 and SII),¹⁹¹ may be investigated.

The data from various laboratories, including our own, show considerable similarities between cortical neural responses during natural and artificial inputs. One could also conclude that the VPL may be an equally qualified or a more suitable candidate for SSNP applications.

Microstimulation has to be applied with caution, however, especially in brain structures that are crucial, such as the thalamus. Leakage into neighboring or interconnected networks and lesions can produce serious adverse responses. One helpful approach could be understanding network dynamics and predicting and optimizing outcomes of microstimulation. Other stimulation methods, such as optical stimulation, that may overcome some limitations have to be tested in terms of SSNP.

Although there are multiple benefits of brain microstimulation with a view toward achieving an optimum BMI, its application in an ethical and safe environment is of concern.⁶⁵ When microstimulation is viable, general guidelines for postoperative care have to be outlined. It has been suggested that brain stimulation could alter a person's brain function and hence his or her personal identity,¹⁴⁹ an issue that needs to be examined from moral, ethical, and practical perspectives.

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